Scalability & Future Plan

Date: 1 November 2026 | Team ID: Not provided — enter the official assigned Team ID before submission | Project: F1 Race Intelligence: Formula 1 Performance Analytics using Tableau

Current system limitations

1. Curated workbook provenance is not verified — impacts historical claims — Priority High.

2. Tableau view inventory/calculated fields require workbook inspection — impacts exact documentation — Priority High.

3. No automated refresh or Snowflake deployment is verified — impacts repeatability at larger scale — Priority Medium.

4. External Tableau/GitHub Pages availability is outside project control — Priority Medium.

Scalability plan

User load: static GitHub Pages is suitable for a read-only project page; future upgrade is CDN/caching or a dedicated application only if needed.

Data storage: current source is Excel; future upgrade is versioned CSV/Parquet and governed warehouse tables with documented schema.

Performance: simplify heavy Tableau marks, use extracts/aggregated views where appropriate, and measure actual load/query times.

Security: keep published data non-sensitive, review licensing, avoid credentials in the repository and use read-only public links.

Future roadmap

Phase 2 — Data and visualization audit: verify source provenance, chart inventory, calculated fields and screenshots; improve glossary and QA checks.

Phase 3 — Reproducible data pipeline: create staging/curated tables and automated row/null/duplicate checks; use Snowflake only if available and authorized.

Phase 4 — Advanced analysis: consider strategy or prediction features only with validated data, transparent assumptions, evaluation and clear separation from descriptive analytics.

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Tableau: https://public.tableau.com/app/profile/sudhanshu.singh3051/viz/Book1_17936408624670/Story1?publish=yes

Website: https://kmtp-png.github.io/Formula1-Race-Intelligence/#tableau